TOOL DESIGN

Subject Code : 23A03504a (Professional Elective - IV)

UNIT IV : DESIGN OF PRESS TOOL DIES

Syllabus (as given)

Types of Dies - Method of Die operation - Clearance and cutting force calculations - Blanking and Piercing die design - Pilots - Strippers and pressure pads - Presswork materials - Centre of pressure - Strip layout - Short-run tooling for Piercing - Bending dies - Drawing dies - Design and drafting.

1. Introduction to Press Working

- *Press working* = shaping sheet metal (cold) by a *press* using a *punch and die set* - cutting or forming, in one stroke.
- Advantages: high production rate (10 to 1000+ strokes/min), low cost per piece, good accuracy and interchangeability, little scrap with good strip layout, semi-skilled labour.
- *Press tool (die set) parts:* upper shoe (punch holder), lower shoe (die shoe), punch, die block, stripper, guide pillars and bushes, shank, pilots, stops, back-up plates, springs, fasteners and dowels.

1.1 Press operations

Group	Operations
Cutting (shearing)	*blanking* (the cut-out piece is the part), *piercing / punching* (the hole is the part), notching, trimming, shaving, parting-off, slitting, lancing, cut-off
Forming	bending, drawing, flanging, curling, embossing, coining, stretching, bulging, extruding, forming (swaging)

- *Blanking* - slug is the product, sheet is scrap. *Piercing* - slug is scrap, sheet with holes is the product.
- *Lancing:* a partial cut and bending without removing metal. *Trimming:* cutting excess

from a drawn/formed part. Shaving: removing a thin layer to get a smooth accurate edge. Notching: cutting from the edge.

2. Types of Dies

2.1 Classification by operation

1. Simple (single-operation) die: one operation per stroke (blanking or piercing only).
2. Compound die: two or more cutting operations at the same station in a single stroke , e.g. blank + pierce a washer; the blank is cut upside down (inverted) - very accurate (concentric) but costly, slower.
3. Combination die: a cutting plus a forming operation at the same station in one stroke (e.g. blank & draw).
4. Progressive (follow-on) die: several operations at successive stations ; the strip advances one pitch per stroke; finished part at the last station . Highest productivity; pilots needed for accurate registering.
5. Transfer die: the part is cut first, then transferred mechanically from station to station for forming (for large parts, deep drawing).
6. Multiple (gang) die / sub-press die for small parts.

Type	Merits	Limitations
Simple	cheap, easy, short lead time	only one operation, low output
Compound	accurate, flat parts, single set-up	expensive, more difficult, slower than progressive
Progressive	high speed, automatic, one man for several ops	high tool cost, big strip needed, high skills to maintain
Transfer	large parts, deep draws	transfer mechanism costly

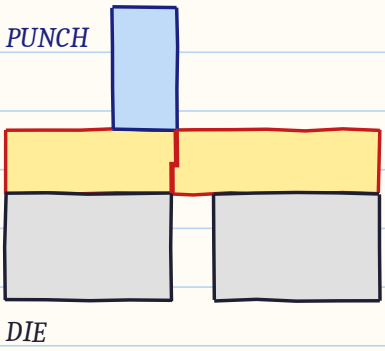
2.2 Classification by construction

- Inverted die: die block on the upper shoe, punch on the lower; the blank is pushed down by a knock-out (flat blanks, compound dies).
- Pillar (guide-post) die sets: two or four guide pillars and bushes keep alignment of punch and die (diagonal, back-post, centre-post). Plain (open) dies: no guide - the press ram guides (cheap, low accuracy).
- Sub-press die: a small die set with guide that is placed in the press; used for small precision parts.

3. Method of Die Operation

- Shearing occurs in three stages as the punch descends into the sheet on the die:

1. Plastic deformation - sheet bends and flows into the die opening; edge shows rollover .
2. Penetration (shearing) - punch cuts about 1/3 (for soft) to 1/4 of the thickness (hard materials); smooth burnish band formed.
3. Fracture - cracks start from the cutting edges of punch and die, meet, and the slug is separated; rough fractured zone and burr .

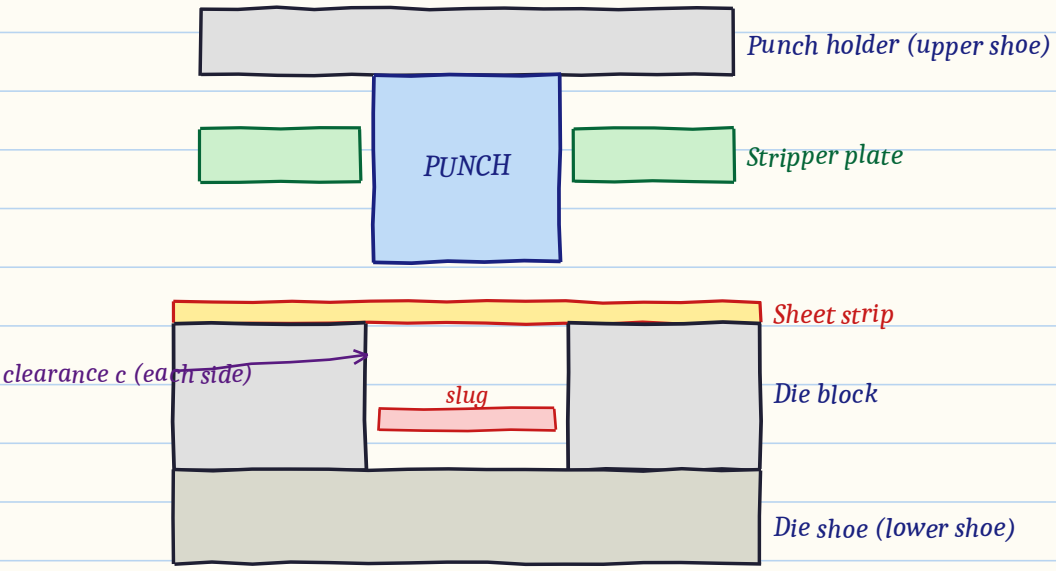


Cut edge of blank shows 4 zones:

1. Rollover (burnish rounding)
2. Burnish (clean sheared band)
3. Fracture zone (rough)
4. Burr (on die side of the blank)

Too little clearance -> secondary shear, high force
Too much clearance -> large burr / rollover

- With the correct clearance , the cracks meet, giving the minimum cutting force and a clean edge .
- Slug goes through the die (relief angle 0.25-2 deg or straight land 3-10 mm then relief of 1-2 deg per side).
- Die land (straight portion): 3-10 mm; after wear re-grind; angular clearance 0.5-2 deg per side.
- Strip is stripped by the stripper (spring or fixed) from the punch on the return stroke, since the sheet grips the punch after cutting (shrinks on the punch).



Blanking die: slug (blank) falls through die opening; strip scrap is stripped from punch

4. Clearance and Cutting Force

4.1 Clearance

- Die clearance c = gap between punch and die opening, per side . It depends on thickness t, material strength and quality of edge desired .

Clearance (per side)

- c = 0.0032 x t x sqrt(tau_s) (mm) ... tau_s = shear strength in N/mm2 (t in mm)
- Thumb rules: c = 5 % t (soft aluminium, brass), 7 - 10 % t (mild steel), 10 - 12 % t (hard steel, stainless)
- Thin sheet (t < 1 mm): c ~ 3-5 % t; Clearance on both sides = 2c

- Too little -> double shear (secondary burnish), higher force, faster punch wear , more stripping force.
- Too much -> large burr, rollover , tapered slug, less accuracy.

4.2 Punch and die sizes (applying clearance)

- Blanking: the blank size = die size (die opening made to blank size); punch = die opening - 2c (clearance is taken on the punch).
- Piercing: the hole size = punch size (punch made to hole size); die opening = punch + 2c (clearance taken on the die).
- Why? Cut edge is the part of the sheet that touches the sharper tool: for a blank it's the die edge; for a hole it's the punch edge.

- Allowance for wear: add (blank) or reduce (hole) by one fourth of tolerance (to start at the worst side of the tolerance) so that the tool wears into the limit.
- Tolerances on punch & die: 5-10 % of the part tolerance (e.g. IT6 / IT7).

4.3 Cutting force

Force formulas

- Cutting force: F = L x t x tau_s (L = length of cut edge = perimeter) [N]
- Shear strength tau_s ~ 0.7-0.8 x ultimate tensile strength
- Reduced with shear on punch/die (slope or taper): F_shear = F x (k t / s) ~ (0.3 to 0.6) F, shear s = 1 to 2 t
- Stripping force = 2.5 to 20 % of cutting force (average 10 %)
- Work done: W = F x t x k (k = 0.4 - 0.6 penetration factor); press power -> tonnage with 20-30 % extra margin
- Press capacity (tonnes) = 1.3 x F / 9810 (safety margin 30 %)

- Example: blank a washer OD 50 mm, ID 20 mm from t = 2 mm mild steel strip, tau_s = 300 N/mm2.
  - L = pi (50 + 20) = 220 mm; F = L t tau = 220 x 2 x 300 = 132 000 N = 13.5 tonne
  - Stripping force 10 % = 1.35 t; press capacity needed = 1.3 x 13.5 ~ 18 tonne -> choose 20 t press.
  - Clearance c = 0.0032 x 2 x sqrt(300) = 0.11 mm per side.
- Reducing cutting force: (1) provide shear (inclined cutting edge) on the punch or die; (2) stagger punch lengths in multiple-punch tools, (3) use a heavy flywheel press.
  - Shear on punch for piercing (die flat) and shear on die for blanking (the punch face flat) so as to keep the part flat . Shear = 1 - 2 t.

5. Blanking and Piercing Die Design

5.1 Die block (die plate)

- Material: tool steel (D2, D3, O1 or T105 / HCHCr), hardened 58-62 HRC; carbide inserts for very long runs; mild steel or cast iron blocks with inserts for large dies.
- Thickness H (thumb rule): H = 0.4 x (cube root of F in N), minimum 12-15 mm; usually 20-40 mm. Thin die for small force, thicker for thick sheets.
- Edge distance (wall): distance from the opening to the edge of the die block = 1.5 to 2 x H (min. 25-30 mm); distance between two openings >= 1.5 H.
- Screws and dowels: at least 2 dowels (diagonal) and 4 socket screws , placed 1.5 x dia.

from the edge - screws M8-M12 spaced about 60-100 mm; keep them clear of the opening .

- Die opening: straight land 3-10 mm then a relief of 0.5-2 deg per side (or 1.5 mm-step relief) so the slug falls free; for sheet thickness less than 1.5 mm, a land = 3 mm.
- Sectional dies (die in pieces) for large, complex shapes: ease of heat treatment, repair and grinding; join lines at corners avoided.

5.2 Punch design

- Material: HCHCr / D2 / HSS, 58-62 HRC; shank (head) softer (30-45 HRC).
- Length of punch: L = (stripper plate thickness + back-up + guiding + (1 to 1.5 t) penetration + 2-5 mm).
- Fixing: (a) head (shoulder) type with retaining plate; (b) screw (plain punch with counter-bored screws); (c) dowel / pins ; (d) shank held in a punch holder; (e) potted with low-melting alloy / epoxy .
- Buckling check of a slender piercing punch (hole dia d < 1.5 t, or d < t for thin punches):
  - Critical load (Euler) Pcr = pi2 E I / (k L)2 ; I = pi d4 / 64; end condition k = 0.5 to 1.
  - Safe if cutting force F = pi d t tau < Pcr / FOS. Piercing diameter d_min ~ 1.0 t (steel) to 0.8 t (soft) without a guide; use a guide (stripper bush) for small punches.
- Punch face: flat, or stepped (staggered) / sheared by 1-2 t to reduce the force and noise; piercing punches with a bulleted nose to hold the slug (slug retaining by a slight dimple/groove).
- Back-up plate (pressure plate): hardened 45-50 HRC; when the punch pressure on the punch-holder > 150-200 N/mm2 , to prevent the punch head sinking into the holder.

5.3 Pilots

- Pilots are small pins that locate the strip precisely in the die before the punch cuts, by entering already-pierced holes - in progressive dies (indirect pilots in the scrap) and compound dies .
- Types: direct pilot (enters a hole that is part of the product - small, 'bullet nose'), indirect pilot (enters a hole punched in the scrap web for registering only).
- Design points:
  - Pilot diameter 0.05 - 0.15 mm smaller than the pierced hole (the hole is placed with a precision of +/- 0.02-0.05 mm).
  - Pilot length = 1 to 1.5 t beyond stripper surface; nose is bullet / conical (taper 30 deg) to move the strip smoothly.

- Pilot must enter before the punch cuts (set 1-2 t longer than punches) and the strip must be free to float (lifted by lifters /guides).
- Pilot head is spring loaded (pressure) or fixed ; pilot is hardened (55-60 HRC) HSS ; replaceable, pilot hole size same as punch/die size set.

- Stock lifters and stops: finger stop, trigger stop (hand-held), automatic stop, side-cutter (pitch notch) , start (stock) stop for first piece, guide rails (strip guide gap = strip width + 0.1 to 0.5 mm).

6. Strippers and Pressure Pads

- Stripper: a plate that holds the strip down and strips it from the punch on the up stroke. It also guides small punches .

Type	Description	Use
Fixed (channel) stripper	a rigid plate fixed on the die with a slot for strip; gap = t + 0.1-0.3; height of slot = 1.5 t + 1 mm	simple; low cost; thin to medium strips; cannot hold work flat
Spring (movable) stripper	plate spring-loaded on the punch holder; presses on sheet before punch cuts	keeps the blank flat; cleaner edge; guides punches; for thin sheets
Positive (knock-out) stripper	rod-actuated or cam-actuated	where spring force is not enough
Pressure pad / knock-out	spring or rubber loaded pad in the die that ejects the blank (inverted dies)	flat blanks; compound dies
Pneumatic / rubber pad	compressed air or urethane	large dies, big forces

- Stripping force: F_s = 2.5 % - 20 % of cutting force (usually 10 %).
- Spring selection: force per spring = F_s / n; stroke of spring = t + penetration + grinding allowance (5-6 mm); use die springs (round wire, rectangular/ heavy) pre-loaded ; or polyurethane/ rubber (compressed 15-25 %).
- Rubber stripping pads: area x pressure; polyurethane pad of 1/3 stroke.
- Stripper plate thickness = 12-20 mm or 0.2 x the strip width , hardened (guided punches) or mild steel; distance from punch to hole in stripper = 0.05 - 0.1 mm (sliding fit) .

6.1 Knock-outs / ejectors

- In inverted and compound dies the blank stays in the die or around the punch -> knock-out (ejector pad) pushes it out by press knock-out bar and pins , spring or air .

7. Presswork Materials

- Press-work stocks should be ductile (high elongation), uniform and with good surface.

Material	Properties	Typical products
Low carbon steel (CRCA, deep drawing steel, IS 513 - D, DD, EDD)	ductile, cheap, good formability	automobile panels, brackets, washers, covers
Stainless steel (304, 316)	high strength, work-hardens	sinks, utensils, medical parts
Aluminium and alloys (1100, 3003, 5052)	light, soft, corrosion resistant	cans, utensils, aerospace, panels
Brass (70/30 cartridge brass), copper	very ductile; high conductivity	electrical terminals, cartridge cases, decoration
Spring steel / phosphor bronze / Be-Cu	high elastic limit	springs, clips, contacts
Non-metals: paper, fibre, leather, rubber, plastic films	cut with steel-rule (cutting) dies	gaskets, insulation, shims

- Stock supplied as sheet, strip or coil ; thickness tolerance +/- 5 % , hardness: dead soft, quarter-hard ... full-hard (bending needs soft, blanking prefers semi-hard for clean cut).
- Shear strength of mild steel ~ 300-350 N/mm2; aluminium 70-100; brass 220-300; stainless 450-500; copper 150-200 N/mm2.
- Grain direction (rolling direction): bends should be across the grain (perpendicular) to avoid cracks; blank orientation considered in strip layout.

8. Centre of Pressure

- *Centre of pressure:* the point where the resultant of all cutting forces acts - the press ram axis (shank axis) must pass through it ; otherwise eccentric loading -> tilting, uneven punch wear, ram guide wear, poor accuracy, breakage .
- *Calculation:* treat each cutting edge by its length (cutting force proportional to the length of the cut edge for the same thickness and material): take moments of the line lengths about two axes .

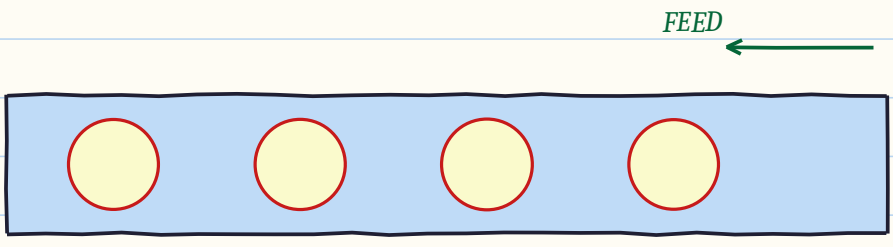
Formulae

- $x = (L1\ x1 + L2\ x2 + L3\ x3 + ...)/(L1 + L2 + L3 + ...)$
- $y = (L1\ y1 + L2\ y2 + L3\ y3 + ...)/(L1 + L2 + L3 + ...)$
- L = length of each cut (circle -> pi d, rectangle -> 2(a + b), line segments - their length); (x, y) its centre (centroid) from the chosen reference axes

- *Example:* piercing 2 holes: hole A dia 10 mm at (20, 20); hole B dia 20 mm at (60, 20); perimeter L1 = 31.4, L2 = 62.8.
  - x = (31.4 x 20 + 62.8 x 60)/(94.2) = (628 + 3768)/94.2 = 46.7 mm ; y = 20 mm . Hence the shank is located at (46.7, 20) - near the bigger hole.
- *Graphical method:* represent each force by a length and find the point with a funicular polygon ; also for non-symmetrical parts the shank is shifted . Put the die set so that the centre of pressure coincides with the shank .
- For progressive dies: use the total force at all stations in a stroke (including the stripping), and consider the forming forces also.

9. Strip Layout

- *Strip layout* = arrangement of the blanks on the strip so as to get maximum economy of material (minimum scrap) with adequate strength of the strip and die .



Strip layout

a = 0.8 t to 1.5 t (min. ~ 1.5 mm)
b ~ 1.5 t to 2 t
Strip width = D + 2b + clearance
Material utilisation = (blank area x blanks) / (strip area used) x 100

9.1 Design points

- Bridge (web) a between blanks and margin b on edges:

Thickness t	Web / margin
Up to 0.8 mm	a = 0.8 - 1.0 mm
0.8 - 3.2 mm	a = t to 1.2 t (min. 1.0 - 1.5 mm); b = 1.5 t
Over 3.2 mm	a = 0.8 t ; b = 1.2 t

- Strip width W = D + 2b + c (c = side clearance for guide 0.2 - 0.5 mm); pitch (feed) P = L + a (L = length of blank in feed direction).
- Material utilisation (%) = (area of blank x blanks per stroke) / (width x pitch) x 100. For good layout 70 - 85 %.

Example

- Circular blank D = 40 mm, t = 1.5 mm; a = 1.5 mm, b = 2.25 mm
- W = 40 + 2 x 2.25 = 44.5 mm; P = 40 + 1.5 = 41.5 mm
- Utilisation = (pi/4 x 40 x 40) / (44.5 x 41.5) = 1256.6/1846.8 = 68 %

- Layouts: single row, double row / staggered (blanks nested: better utilisation), angled (for L,T shapes), multiple row , and scrapless (parting, cut-off dies, no waste).
- Provide pilot holes in scrap , feed direction , grain direction , and burr side (burr should face away from the visible/ assembly side).
- In progressive dies, sequence of operations: pierce first (the small holes), notching, then forming/bending, finally blank (cut-off) ; idle stations to strengthen die; carrier strip held to the end.

10. Short-run Tooling for Piercing

- For low volumes (prototype, a few hundred to a few thousand pieces) - cheap, quick and flexible tooling is used instead of full die sets.
- Methods:

1. Pierce and blank dies made of cheap material: mild steel or Kirksite (zinc alloy) die with a plastic or low-hardness punch ; epoxy-faced dies with steel punch.
2. Dimple / sub-press dies: simple punch and die plate with a stripper using standard components (punch & die sets from catalogues), with locating gauge plate .
3. Rubber (urethane) pad (Guerin / Marform) piercing: a steel punch and soft rubber pad as a die ; works for thin soft sheet metal and gaskets.

4. Steel-rule (cutting) dies: steel knife edge bent to profile and set in wood/plywood block , rubber ejection pads ; for paper, leather, plastic, rubber, thin soft metals.
5. Turret punch presses (CNC) and nibbling: a standard-shaped punch in turret , no special tool ; the sheet moves on a CNC table ; laser/plasma cutting is also used instead.
6. Universal/ modular piercing units: pre-assembled punch holders on a base plate (like hole-punching units), quick set-up ; standard shape holes (round, square, slot) .
7. 3-D printed polymer or aluminium dies for very short runs.

- Advantages: low cost, quick delivery, easy changes; limits: low accuracy, low life, thin sheets only, slower.

11. Bending Dies

- Bending: plastic deformation of sheet about a straight axis ; outer fibres stretch, inner fibres compress ; the neutral axis shifts toward the inside.
- Types: V-bending (die), U-bending, wiping (edge) bending, air bending, rotary bending, bottoming, coining; channel, offset, curling, hemming, seaming.

11.1 Bend allowance and development

Bend allowance (length of neutral axis)

- BA = (pi / 180) x theta x (R + k t) (theta = bend angle in deg; R = inside radius; t = thickness)
- k = 0.33 (R < 2t), 0.4 (2t < R < 4t), 0.5 (R > 4t)
- Developed length = sum of flat lengths + sum of BA
- Minimum bend radius: soft steel 0.5 t, aluminium 1 t, hard brass 2 t, hard stainless 3 t (across grain) - not less, else cracks

- Springback: after the load is removed, the sheet partially recovers - angle opens by 1-10 deg. Springback is more in high strength material, large radius and thin sheet. Compensation: over-bending (die angle smaller than required by the springback angle), bottoming (coining), stretch bending , radius smaller , pressure pad .
- Springback factor: Ks = (R_i + t/2) / (R_f + t/2), where R_i = radius of the tool and R_f = radius after springback. Ks < 1, so the tool is made with a smaller radius / angle (overbend) to get the required final bend.

11.2 Bending force and die design

- V-die: F = (C x sigma_u x L x t2) / W (C = 1.33; W = die opening = 8 t; L = length of bend)
- U-die (channel): F = (0.67 to 0.8) x sigma_u x L x t2 / W ; wiping die: F = (0.33 sigma_u L

t2) / W, plus pressure pad force (30 - 50 % of F).

- Die design: die radius (entry) 3 - 5 t, punch radius = required inside bend radius (>= min. radius) , die opening width W = 8 t for V, die depth so as to allow the part, clearance in U-die c = t + 0.05 - 0.1 mm (max thickness + allowance), side relief .
- Pressure pad / knock-out and stops (pins or gauge plate) to locate the blank ; pilot pins to prevent the slippage (blank skids when bending).
- Material: tool steel (T 105 / D2) hardened 55-60 HRC; finish polished (Ra 0.2 - 0.4).
- Bending sequence: bend the outside bends first , then inside bends; bend lines at right angle to the grain.

12. Drawing Dies

- Drawing: converts a flat blank into a hollow cup (cylindrical, rectangular or irregular) by pushing a punch into a die; the flange flows radially inward (the sheet is under radial tension and circumferential compression).
- Types: first draw (cupping), redraw (reverse and direct), ironing (thin the wall), deep drawing, stretch forming, hydro-forming ; single-acting, double-acting (blank holder) press.

12.1 Blank development

- Equal surface area principle (thickness unchanged): Area of blank = area of the finished cup (with trim allowance).
- Cup of diameter d and height h (flat bottom, sharp corner): D = sqrt(d2 + 4 d h) .
- With bottom corner radius r: D = sqrt(d2 + 4 d h - 0.5 r d) (approximate; r = bottom corner radius) add 2 - 5 mm trimming allowance on the height.

12.2 Draw ratio and number of draws

Drawing parameters

- Draw ratio DR = D / d (limit 1.8 - 2.0 for first draw for steel); reduction % = (D - d)/D x 100
- First draw: 40 - 50 % ; redraws: 25 - 30 % (second), 15 - 20 % (third) ... with intermediate annealing after 2 - 3 draws
- Drawing force: F = pi d t sigma_u x (D/d - 0.7) (empirically); or F = pi d t sigma_u x C C = 0.6 - 1.0
- Blank holder force: FBH = (1/3) x F or p x A : p = 1.5 - 3 N/mm2 (steel) x area of the flange under the holder
- Press force = F + FBH (+ margin 30 %); Work done = F x h x (0.6 to 0.7)

- Example: cup d = 50 mm, h = 40 mm, t = 1 mm: D = sqrt(2500 + 4 x 50 x 40) = sqrt(10 500) = 102.5 mm ; DR = 2.05 -> 2 draws needed ; first draw d1 = 0.6 x 102.5 = 61.5 mm (40 % reduction), second to d = 50 mm (19 % reduction).

12.3 Drawing die design

- Punch and die corner radii: die radius r_d = 4 to 10 t (soft/thin small, hard/thick large); punch radius r_p = (3 to 4) t ... up to 10 t (maximum = 1/2 of wall). Small die radius -> tearing (fracture) , large -> wrinkling.
- Clearance (per side) c = t + (0.07 to 0.2) x sqrt(10 t) or c = 1.1 t to 1.3 t (steel 1.1 t, aluminium 1.05-1.1 t, stainless 1.2 t); too tight -> ironing, tearing .
- Punch diameter = d (inside of cup) ; die diameter = d + 2t + 2 (c - t) i.e. punch + 2c.
- Blank holder (pressure pad) prevents wrinkles; spring, rubber, pneumatic or double-action press hold; draw beads control the flow.
- Air vent in the punch (hole 3-6 mm) to avoid vacuum when stripping the cup.
- Lubrication: heavy oil, wax, polyethylene film, soap; die material alloy steel / cast iron, chrome plated or nitrided ; aluminium bronze inserts for stainless steel; carbide for long-run; finish Ra 0.1 - 0.2 .
- Defects: wrinkling (flange: low blank holder force, big DR), tearing / fracture (too much force, small radius), earing (anisotropy), surface scratch , springback of walls.

13. Design and Drafting of Press Tools

13.1 Design procedure

1. Study the component drawing - material, thickness, tolerance, quantity, press available.
2. Decide the operations and the sequence ; select type of die .

3. Strip layout -> pitch, strip width, utilisation.
4. Calculate clearance, cutting force, stripping force, press tonnage , centre of pressure.
5. Determine die block, punch, stripper, die set dimensions; select standard die set (IS 12 ... or manufacturer) with guide pillars .
6. Select materials & heat treatment ; fasteners , springs, stops.
7. Check press data: shut height, bed and bolster size, stroke, shank diameter.
8. Prepare assembly and detailed drawings , BOM, and check .

13.2 Drafting practice (what to show)

- Plan (top view) with upper half removed: shows die block, stripper, guide rails, stops, strip layout in chain-line. Elevation: sectional front view showing punch, die, stripper, shoe, pillar, springs, shank in closed position.
- Strip layout and component drawing on the sheet; the die set is in 'open and closed' or shown in section at the 'closed position' .
- Bill of materials: part no., name, quantity, material, heat treatment, standard/remarks. Notes: hardness, tolerances (punch & die clearance given), grinding allowance, dowel/ screw positions , and the shut height .
- Standard die set parts: upper shoe, lower shoe, guide pillar (ground, press-fit H7/p6), guide bush (slip fit H7/h6 or ball-bearing bush), shank (centre of pressure), punch holder, back-up plate, stripper, spacers.

Exam checklist for any die design question

- Write operation, material, thickness, tau ; clearance , punch & die size ; cutting force , stripping , press tonnage
- Draw strip layout (pitch, margin, bridge, utilisation) and centre of pressure
- Sketch the die assembly (plan + section) with names of all parts; give the BOM

14. Quick Revision - Unit IV

- Dies: simple, compound, combination, progressive, transfer. Clearance c = 0.0032 t sqrt(tau); blank -> die size; hole -> punch size .
- F = L t tau; stripping 10 %; press tonnage with 30 % margin; shear on punch/die reduces force.
- Pilots register the strip; stripper (fixed/spring); pressure pad / knock-out.
- Centre of pressure = (sum L x)/(sum L); shank at CP. Strip layout: a = t to 1.5 t, utilisation = blank area/(W x P).

- Bending: BA = (pi/180) theta (R + k t); springback -> overbend. Drawing: D = sqrt(d2 + 4 d h); first draw 40-50 %, die radius 4-10 t, c = 1.1 t.